Language-Guided Spatial Prompting For Part Segmentation Of Generated 3D Assets

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Abstract

Generated three-dimensional (3D) assets often lack the part structure needed for editing. Automatic masks can merge functional components, while language alone cannot locate their boundaries. We introduce language-guided spatial prompting to construct part maps for a frozen 3D segmenter. Qwen3-VL specifies visible parts using names, boxes, and positive and negative points. It audits SAM2 masks, allowing at most one prompt correction and retry per rejected candidate. Deterministic assembly resolves overlaps and fills foreground gaps before the map conditions SegviGen. Controlled experiments share reconstructed geometry, model parameters, and inference settings. Local integer labels preserve part identities during assembly before palette encoding for the backend. Our two-dimensional guidance reaches 53.64% matched mean intersection over union (mIoU) on 479 objects. On the controlled 479-object projection set, mIoU reaches 37.75%, compared with 35.50% for automatic guidance and 33.64% for the raw image. These results measure agreement with designated source-view references and show how guidance construction can improve a frozen 3D backend without retraining.

Keywords: 3D Part Segmentation, Vision-Language Models, Spatial Prompting, Mask Refinement, Generated Assets

1 Introduction

Image-conditioned generation can produce textured 3D assets from a single observation. Compact structured representations now support objects with varied shapes and appearances (Xiang et al., 2026). The generated surface still gives no direct indication of the functional parts a user may want to select, inspect, or edit. An outdoor air-conditioning unit, for example, contains a grille, service panel, and housing even when those surfaces share similar materials. Recovering these regions requires both a definition of the intended parts and spatial evidence for their boundaries.

Part-aware generation offers another route. PartCrafter jointly generates distinct part meshes, while PartGen combines multiview part segmentation with generative completion and reconstruction (Chen et al., 2025; Lin et al., 2025). We hold the reconstructed asset fixed and study the partition supplied to its segmenter. Existing methods learn 3D part features, support text queries, or predict masks directly from native 3D data (Liu et al., 2025; C. Ma et al., 2025; Z. Ma, Yue, &

Gkioxari, 2025; Zhu et al., 2026). Other methods transfer prompted 2D segmentation into a 3D representation (Cen et al., 2025; Deng et al., 2026). SegviGen formulates part segmentation as structured coloring under a pretrained 3D generative prior (Li et al., 2026). Its 2D-guided mode lets us examine how a source-view partition affects the final 3D output. The outcome depends on the partition's granularity, spatial support, and treatment of ambiguous regions.

Automatic image segmentation can produce plausible regions that do not match the parts required for a particular object. Similar appearance may cause a grille and its surrounding panel to share one mask. Another prompt may isolate a narrow frame while missing the larger component to which it belongs. Language identifies the intended component, but the name alone cannot locate its pixels. We therefore pair each part description with a box and point evidence, and allow one correction when the first prediction is wrong. In Fig. 1, automatic guidance merges regions that remain separate in the language-guided result and the designated reference.

Guidance construction combines part planning, per-mask audit, and deterministic assembly. Qwen3-VL predicts names and spatial prompts (Bai et al., 2025), and SAM2 converts them into masks (Ravi et al., 2025). The VLM inspects each named candidate as a red overlay on the source image. A rejected mask receives revised prompts and one retry. Assembly then converts the selected masks into a foreground-complete guidance image. The audit uses source appearance and part names because palette colors carry no intrinsic semantics.

The downstream 3D model remains fixed. The source image also yields a shared reconstructed asset, which is normalized and voxelized for SegviGen. In the controlled A/B/C/D comparison, only the guidance condition changes: A uses the source image, B uses automatic SAM2 guidance, C uses the proposed pipeline, and D uses the author-designated human reference. The comparison measures how 2D guidance changes the final 3D partition under shared geometry.

We contribute a source-view guidance procedure that connects named parts to spatial evidence for frozen 3D propagation. Deterministic assembly retains the original box and positive anchor even when a retry changes the segmentation prompt. Evaluation first compares final-output performance with existing methods, then separates source-view guidance quality from propagation under shared SegviGen settings. Component, feedback, and structural analyses examine where improvements arise and where they fail.

2 Related Work

2.1 Part Representations And 3D Segmentation

PartSLIP lifts image-language detections to point clouds, while PartDistill distills vision-language information for shape part segmentation (Liu et al., 2023; Umam, Yang, Chen, Chuang, & Lin, 2024). Parts2Words aligns point-cloud parts and words for cross-modal retrieval (Tang, Yang, Wu, Han, & Chang, 2023). Learned representations offer several routes to 3D part understanding. Find3D learns features that localize parts from open text queries (Z. Ma et al., 2025), while PartField learns continuous fields for clustering parts at different granularities (Liu et al., 2025). P3-SAM and PartSAM formulate promptable segmentation on native 3D data and include automatic decomposition procedures (C. Ma et al., 2025; Zhu et al., 2026). SAMPart3D produces scale-conditioned decompositions and assigns names afterward, whereas ALIGN-Parts jointly addresses segmentation and naming (Paul, Kaushik, Vaidya, Bhattad, & Yuille, 2025; Yang et al., 2024). In our pipeline, names define spatial prompts before segmentation and remain 2D metadata; they are not evaluated as semantic labels of the final 3D regions. Hierarchical representations further distinguish parts from subparts (Gao, Hu, & Liao, 2025). ADE20K explicitly distinguishes objects, parts, and some nested subparts (B. Zhou et al., 2019). Granularity is therefore an explicit evaluation choice. We specify visible source-view parts before applying an existing 3D model, so the added mechanism operates at the guidance interface.

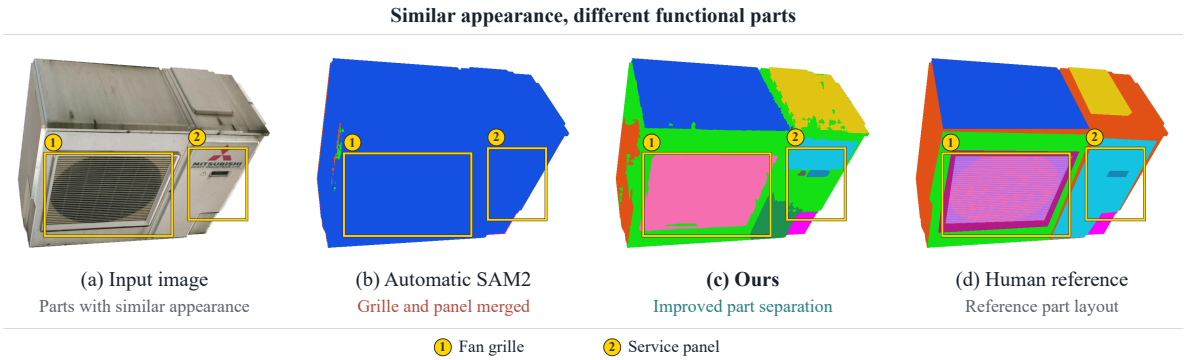


Fig. 1 Motivation: source image, automatic SAM2 guidance, our guidance, and the designated human reference. Yellow boxes highlight the fan grille and service panel, which have similar appearance but distinct functions. Display colors are matched while region boundaries are retained.

2.2 Language And Spatial Evidence For Parts

Text can identify a functional part, while visual evidence locates it. CoOp adapts frozen vision-language backbones through learned context vectors (K. Zhou, Yang, Loy, & Liu, 2022). Our boxes and points are instead generated for each object, without learning additional prompt parameters. DM-PCL studies text-driven consistency for weakly supervised few-shot 2D part segmentation (Han et al., 2025). GMENet models object semantics, part relations, and boundaries through graph matching (Michieli & Zanuttigh, 2022), and cross-model prior fusion combines semantic and spatial representations (Zhao et al., 2026). Earlier scene parsing selects candidate object regions while enforcing consistent overlap and occlusion relations (Tighe, Niethammer, & Lazebnik, 2015). Our pipeline links part names, boxes, and points through deterministic mask assembly. Interactive methods use clicks to select candidates (Jain & Grauman, 2019), and reasoning video segmentation has reused predicted masks as dense prompts (Dang et al., 2025). Here, the audit examines one source-view part overlay at a time and returns box-and-point corrections before deterministic assembly for a frozen 3D model.

2.3 2D Evidence And 3D Propagation

OmniSeg3D learns a consistent hierarchical 3D feature field from multiview 2D segmentation evidence (Ying et al., 2024). Radiance-field segmentation can propagate a prompted 2D mask through inverse rendering and cross-view self-prompting (Cen et al., 2025). GeoSAM2 predicts masks on multiview geometric maps and aggregates them in 3D (Deng et al., 2026). SegviGen uses a 3D generative prior and accepts a 2D part map as the condition for complete segmentation (Li et al., 2026). We adopt the SegviGen interface with fixed parameters, modifying only the construction and audit of the input map. Evaluation studies have also shown the value of separating sources of segmentation difficulty through controlled comparisons (Yang & Yang, 2024). Accordingly, we report guidance, projection, and structural-group results as separate analyses.

3 Method

3.1 Overview And Problem Definition

The input is a single red, green, blue, and alpha (RGBA) rendering I of an object. The same rendering is used for 3D reconstruction and part planning. Let I_{rgb} denote its color channels and let $\alpha(x)$ denote its alpha value at pixel x. The foreground support is

$$F_\alpha = \{x \mid \alpha(x) > 10\}. \tag{1}$$

At the source resolution, the method produces an integer label map L, a color guidance map G, and a metadata record $\mathcal{A}$. SegviGen combines G with the shared voxel representation V to produce a

colored 3D partition. Figure 2 traces the complete data flow; its shaded 3D assets illustrate the interfaces between stages.

The geometry branch supplies a common representation for every guidance condition of an object. The guidance branch changes part assignments without regenerating the geometry. This separation matters because the downloaded asset and its reconstruction may differ in topology and surface detail: the former provides the source rendering, while the latter is the object segmented downstream.

3.2 Shared 3D Asset Construction

TRELLIS.2 generates a textured mesh from I (Xiang et al., 2026). The exported binary glTF file, denoted by $\mathcal{G}$, is an intermediate representation. Geometry preprocessing applies the experiment's fixed normalization and coordinate convention before voxelization. We write this transformation as

$$V = \mathcal{V}\big(\mathcal{N}(\mathcal{G})\big), \tag{2}$$

where $\mathcal{N}$ is the configured geometric preprocessing and $\mathcal{V}$ produces the sparse voxel asset stored as VXZ. SegviGen receives the resulting voxel representation together with the source-view guidance.

The collection uses upright source renderings with orientation fixed before guidance generation. A/B/C/D runs reuse each object's reconstructed mesh and VXZ, along with the same segmentation checkpoint, sampling configuration, and random seed. Their output partitions are therefore compared under one 3D setup. The reproduction record also stores the preprocessing transform and source camera, because silhouette agreement alone cannot verify correspondence between source-view labels and rendered 3D surfaces.

3.3 Visible Part Planning And Spatial Prompts

Qwen3-VL receives I_{rgb} and proposes the visible components to be segmented (Bai et al., 2025). Each candidate is represented by

$$r_i = (n_i, b_i, P_i^+, P_i^-), \qquad i = 1, \ldots, K, \tag{3}$$

where n_i is the part name, b_i is a bounding box, and P_i^+ and P_i^- are positive and negative point sets. The proposal specifies a component visible in the given view. Repeated components can retain the same name while using different spatial records. This keeps semantic category and instance identity separate.

The box limits the search region. Positive points provide interior evidence, while negative points exclude nearby components that could otherwise be merged. The planner predicts coordinates in $[0, 1000]$, which are converted to image coordinates using the actual width and height. Parsed values are clipped to valid image coordinates. Positive points are checked against the foreground and adjusted within the supported neighborhood when necessary. A proposal with no valid positive support is recorded and skipped. The part name remains associated with the record, but is not passed to SAM2 as a text prompt.

For every valid proposal, frozen SAM2 predicts a single candidate mask

$$M_i^{(0)} = S_\theta(I_{\text{rgb}}, b_i, P_i^+, P_i^-). \tag{4}$$

SAM2 uses all valid positive and negative points and returns one mask per proposal. When available, its predicted quality score is stored as model confidence, independently of the evaluation reference. Multiple positive points can support separated visible portions of one component, while negative points distinguish nearby surfaces inside the same box. The prompt ablations remove these signals independently from both the initial prediction and the retry.

3.4 Per-Part Audit And One-Pass Refinement

Before merging, the VLM inspects every initially valid mask. It receives the source image, candidate name, and a semitransparent red mask overlay with a two-pixel contour. The source appearance helps distinguish a missing interior region from a correctly excluded neighboring surface, a distinction that a binary silhouette may conceal. Each audit remains attached to one candidate, so a rejected region can be traced to its spatial prompt. Figure 3a illustrates the audit and single retry.

An accepted candidate retains $M_i^{(0)}$. A rejected candidate receives a corrected box when provided and additional positive or negative

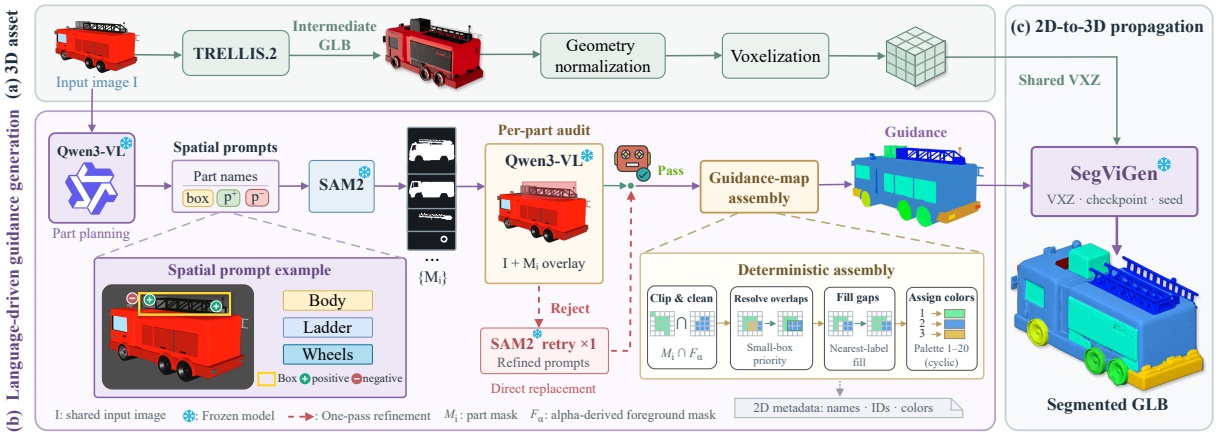


Fig. 2 Pipeline overview. The upper branch reconstructs and voxelizes a shared 3D asset. The lower branch plans parts, predicts masks, audits candidates with at most one retry, and assembles a complete guidance map. The shared VXZ and guidance condition frozen SegviGen; names and local IDs remain 2D metadata. Asset and mask drawings are schematic.

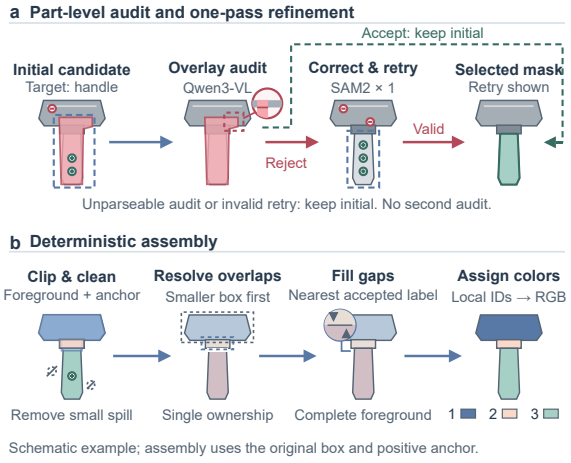


Fig. 3 Part-level audit and assembly, illustrated with a schematic hammer. **a**, Circled signs mark positive/negative points; dashed boxes mark spatial prompts. The zoom highlights mask spill. Accepted masks are retained; rejection permits one correction and retry. Unparseable audits or invalid retries retain the initial mask, without a second audit. **b**, The original positive anchor guides cleaning, smaller original boxes receive overlap priority, and gaps inherit the nearest accepted label. Integer IDs precede palette encoding.

points. SAM2 is called once more with the resulting prompt

$$M_i^{(1)} = S_\theta\big(I_{\mathrm{rgb}}, \widetilde{b}_i, P_i^+ \cup \Delta P_i^+, P_i^- \cup \Delta P_i^-\big). \quad (5)$$

A valid retry directly replaces the initial candidate. The procedure performs no second audit and does not choose between old and new masks by overlap. If the audit cannot be parsed or the retry is invalid, it retains the initial valid mask and records a partial event. Every proposed part keeps its audit outcome, and the number of segmenter calls remains bounded.

Refinement changes the retry prompt, while assembly continues to use the initial proposal's box and first valid positive point for overlap priority and component anchoring. Mask correction thus remains separate from the merge order. Let $N \le K$ denote the proposals entering the initial SAM2 call and $K_v \le N$ those with valid initial masks. The procedure makes N initial calls, K_v audits, and at most K_v retries, so SAM2 is called no more than $N + K_v \le 2N$ times. This bound counts calls and does not estimate wall-clock time.

3.5 Deterministic Guidance-Map Assembly

Individual masks may overlap, contain disconnected spill, or leave foreground pixels uncovered. Assembly converts them into a complete partition without asking the VLM to choose RGB values. Figure 3b illustrates the four operations. Each semantic proposal retains a distinct integer label until the resolved partition is converted to palette colors.

3.5.1 Clip And Clean

Let M_i denote the mask selected by the audit procedure and let p_i denote the first valid positive point of the initial proposal. The cleaned

candidate is

$$\overline{M}_i = \mathcal{C}(M_i \cap F_\alpha, p_i). \tag{6}$$

The cleaning operator first finds eight-connected components. It keeps the component containing p_i as the main component. If the point lies outside the predicted region, the nearest component is used; if no anchor is available, the largest component is selected. Additional components are retained when their area is at least $\max(96, \lfloor 0.18 a_i \rfloor)$, where a_i is the main component's area. This rule preserves separated visible fragments while suppressing small spill regions.

The mask then undergoes closing with a 3×3 elliptical element, another foreground clip, and component filtering. Small enclosed holes are filled if their area is at most $\max(48, \lfloor 0.00015 |F_\alpha| \rfloor)$. Background regions connected to the image boundary are not filled as holes. A cleaned candidate must meet the area threshold $\max(a_{\min}, 96, \lfloor 0.00012 |F_\alpha| \rfloor)$, where $a_{\min}$ is the configured minimum. The default implementation uses $a_{\min} = 10$ at the source resolution. These are resolution-dependent image operations.

3.5.2 Resolve Overlaps

Large masks may include both a small component and its surrounding housing. Letting the larger mask claim the overlap first can erase the smaller component. Assembly therefore sorts candidates by increasing initial box area, with cleaned mask area and candidate ID as deterministic tie breakers. With order π and initially empty occupied set U_0, the remaining region of the next candidate is

$$R_{\pi(t)} = \overline{M}_{\pi(t)} \setminus U_{t-1}. \tag{7}$$

The region is accepted if it has at least $\max(a_{\min}, 96, \lfloor 0.00010 |F_\alpha| \rfloor)$ pixels. Its pixels receive a new local ID and enter the occupied set. Otherwise the candidate is dropped with an explicit reason. Box area determines priority independently of the part name and the evaluation reference. The earlier box-and-point baseline uses a different ownership rule together with other assembly changes, so its comparison does not isolate small-box priority alone.

3.5.3 Fill Gaps

Let U be the union of accepted regions. Every unassigned foreground pixel is given the label of its nearest accepted pixel

$$L(x) = L\left(\arg\min_{y \in U} \|x - y\|_2 \right), \qquad x \in F_\alpha \setminus U. \tag{8}$$

A Euclidean distance transform implements this assignment. Existing labeled pixels retain their ownership, background pixels remain zero, and gap filling introduces no new semantic proposal. The procedure records a failure if no candidate survives.

When at least one candidate survives, the final integer regions are mutually exclusive and their union equals F_α. Subtracting occupied pixels before assignment and filling only unassigned foreground guarantees this well-formed condition. Foreground completeness says nothing about whether the chosen parts or boundaries are correct, so evaluation also requires region matching and part-count metrics.

3.5.4 Assign Colors

The integer map is saved before palette conversion. With palette $c_0, \ldots, c_{20}$ and background index zero, the guidance color is

$$G(x) = \begin{cases} c_0, & L(x) = 0, \\ c_{1+((L(x)-1) \bmod 20)}, & L(x) > 0. \end{cases} \tag{9}$$

The script assigns colors deterministically; the VLM neither chooses palette indices nor edits the assembled image. When a map contains more than 20 labels, the foreground colors repeat cyclically. The 16-bit label image remains the identity-bearing representation because one RGB value may then represent several local IDs. Metadata stores the retained name, local ID, palette index, and associated mask for each 2D region.

3.6 Frozen 3D Propagation

The downstream interface is summarized by

$$\widehat{\mathcal{G}}_k = H_\phi(V, G_k; \xi), \qquad k \in \{A, B, C, D\}, \tag{10}$$

where ϕ denotes the frozen SegviGen checkpoint and ξ its sampling configuration. Condition

A uses the source image as G_A instead of an explicit part map. B uses automatic SAM2 guidance, C uses our assembled guidance, and D uses the designated human reference map. SegviGen extracts the image condition, samples its structured texture representation, and exports the colored result. The method figure abstracts these internal stages to focus on construction of the upstream condition.

Part names remain in separate 2D metadata and are never supplied to the 3D model as text. Colors in the exported GLB are also not persistent semantic surface IDs; such IDs would require an explicit surface-to-label correspondence. We evaluate the colored partitions with the label and projection protocols below, without treating the 2D metadata as semantic labels for individual 3D surfaces.

4 Experiments

The experiments first establish final-output performance relative to existing methods, then examine the contribution of guidance under a fixed 3D backend. We evaluate the source-view maps before their propagated outputs, study prompting, assembly, and feedback, and then examine structural variation and failure cases. The feedback and part-size analyses retain their earlier cohorts and protocols, distinct from the primary comparison.

4.1 Data And Evaluation Sets

The assets come from Objaverse (Deitke et al., 2023). The final author-selected roster contains 500 unique objects: 157 assemblies, 181 solid-volume objects, and 162 thin or fine-grained objects. These groups describe structural difficulty, not semantic classes. The controlled projection comparison uses 479 objects jointly evaluated under A/B/C/D: 151, 179, and 149 in the respective groups. Direct 2D guidance scores are available for C on all 479 objects and B on 440; 39 B predictions are missing.

An earlier 500-object manifest supports the archived feedback and part-size analyses. It contains 154 assemblies, 181 solid-volume objects, and 165 thin or fine-grained objects, but its UID membership differs from the final roster. Sections 4.8 and 4.9.3 retain the original reference and aggregation protocols for those analyses. Their means are excluded from the primary comparison.

Asset availability and successful C runs determined the retained collection, so paired quality comparisons are conditional on that selection. Missing inputs, failed inferences, and evaluation-gate exclusions are recorded separately instead of being assigned zero segmentation scores. Computing runtime failure rates would require the full attempted population before filtering. *[Finalize the development/test split and attach the attempted-population manifest with exclusion reasons.]*

D is the author-designated source-view reference derived from human-repaired part maps. Integer IDs and colors are local to each object, and the mapping files retain provenance and verification status. All reference-readiness flags are false in the archived 500-object feedback analysis. Its scores therefore measure consistency with the specified reference rather than agreement with independently verified ground truth. *[Complete the reference audit and report annotation granularity, ignored regions, and inter-annotator checks.]* The reference labels visible image regions, not a complete 3D mesh. Condition D supplies this same partition to the 3D model and uses it to score the projection, so D measures reference-conditioned consistency instead of an independent blind-test upper bound.

4.2 Implementation And Controlled Conditions

Qwen3-VL performs planning and mask audit, while SAM2 predicts prompt-conditioned masks. Source images and guidance maps are 1024×1024. The available five-object configuration uses Qwen3-VL-4B-Instruct and SAM2.1 Hiera base-plus. *[Confirm the model identifiers and frozen configuration for the full C run.]* All pretrained components remain frozen, and each object's guidance construction uses the same prompt parsing, source-image orientation, and alpha support.

The recorded 3D configuration uses voxel resolution 512, the `full_seg_w_2d_map.ckpt` checkpoint, 20 sampling steps, and seed 42. The export configuration enables remeshing with a target of 100,000 faces and a texture size of 4096. The

shared geometry and sampling configuration are common to A/B/C/D for each retained object. The manifest links each guidance condition to its actual input geometry and inference configuration.

4.3 Comparison Methods

The external comparison includes Find3D, PartField, P3-SAM, GeoSAM2, and SegviGen (Deng et al., 2026; Li et al., 2026; Liu et al., 2025; C. Ma et al., 2025; Z. Ma et al., 2025). Together they cover text-conditioned features, part clustering, automatic decomposition, multiview prompting, and generative segmentation. Table 1 records the information available to each method. PartField uses the recorded $K = 10$ clustering setting, and P3-SAM uses automatic decomposition. We report every competing result because region agreement, boundary agreement, and part recovery can yield different rankings.

GeoSAM2 uses spatial prompts and multiview geometric maps, whereas Find3D requires part text queries. The controlled A/B/C/D experiment differs by changing only the condition supplied to a shared SegviGen backend. Table 1 reports results from our evaluation protocol and excludes benchmark scores from other datasets. *[Attach the frozen external-method configurations, including Find3D query construction and the GeoSAM2 shared-prompt adaptation.]* The 479-object rows form the primary paired comparison. Find3D is reported separately because its retained summary covers 468 objects and uses a different aggregation.

4.4 Guidance And Projection Metrics

Predicted integer labels are compared with reference regions after excluding the declared ignore values -1 and 65535. Background has label zero and is excluded from the part list, while predicted leakage into reference background remains in the region union. For predicted parts $P_1, \ldots, P_m$ and reference parts $D_1, \ldots, D_n$, intersection over union (IoU) is

$$J_{ij} = \frac{|P_i \cap D_j|}{|P_i \cup D_j|}. \tag{11}$$

Local integer IDs and display RGB values may differ across methods, so region correspondence is determined by overlap rather than palette identity.

Matched mean IoU (mIoU) uses a one-to-one assignment that maximizes the sum of pairwise overlaps

$$\mathrm{mIoU}_{\mathrm{match}} = \frac{1}{n} \max_{\pi} \sum_{j \in \mathrm{dom}(\pi)} J_{\pi(j),j}. \tag{12}$$

Here π is a partial injection from reference to predicted parts. The fixed denominator n assigns zero contribution to unmatched reference regions. Part F1 evaluates additional predictions at an IoU threshold of 0.5. Its matching maximizes the number of pairs that meet the threshold; with T matched pairs, the score is $2T/(m+n)$. Because this objective differs from maximizing the IoU sum, the two scores reveal different forms of missed or excessive partitioning.

Boundary F1 (BF1) uses nonzero-overlap pairs from the IoU assignment at 1024×1024. Per-pair values are divided by the reference part count, with zero contribution from unmatched reference regions. Ignored pixels follow the metric implementation's boundary treatment. We distinguish five- and two-pixel tolerances explicitly. Both tolerances use identical predictions and references within each reported cohort. The component table marks BF1 entries whose five-pixel provenance remains unverified.

Each metric is computed per object. Object-weighted summaries average over the stated UID set; structural-group macro-averages weight the three group means equally. The primary projection comparison uses 479 objects. Its additional two-pixel BF1 column covers 471 objects for four methods. Direct 2D summaries use the available B and C sets and are not paired comparisons. Component and archived feedback tables retain group-equal macro-averaging.

For final-output evaluation, each segmented GLB is projected through the frozen source camera into a discrete label map. Label rendering avoids illumination-dependent colors. Before scoring, we verify asset identity, camera provenance, foreground preservation in every arm, and face-count preservation. The controlled A/B/C/D protocol requires camera-to-source foreground IoU

of at least 0.50 and all six pairwise foreground IoUs of at least 0.99. Failure of either shared gate removes the object from every arm. Scoring uses the original projected label maps without intersection, resizing, or pixel repair.

Projection scores measure agreement between the final 3D partition and the visible source-view reference under the frozen camera. They depend on both part assignment and the reconstructed surface visible from that camera, and do not cover unobserved surfaces or volumetric 3D IoU. The qualitative figures use separately recorded display views, which are not treated as quantitative evidence.

4.5 Comparison With Existing Methods

Table 1 compares final-output projection agreement on the common 479-object set; Find3D retains its original coverage and is reported separately. Ours obtains 37.75% matched mIoU and 35.58% BF1. P3-SAM has higher matched mIoU (39.83%) and Part F1 (37.04%), while Ours has higher BF1 than P3-SAM (32.67%). The ranking therefore depends on whether the metric emphasizes region overlap, boundaries, or part recovery. Relative to the shared SegviGen raw-image condition, our guidance raises mIoU by 4.11 points and BF1 by 4.36 points. Figure 4 shows the metric profiles and separates Find3D because of its different coverage.

Figure 5 presents six favorable cases selected to examine where the additional guidance helps. Ours separates the mug handle from the vessel, the bat grip from its body, and the fan support from its base. The same examples also expose residual fragments and differences in part granularity. These selected cases illustrate local gains rather than the aggregate ranking in Table 1.

4.6 Effect Of Guidance On The Frozen Backend

The controlled conditions examine the guidance interface in two stages. Direct 2D evaluation measures the supplied partition, while output projection evaluates its downstream effect under shared geometry and SegviGen settings.

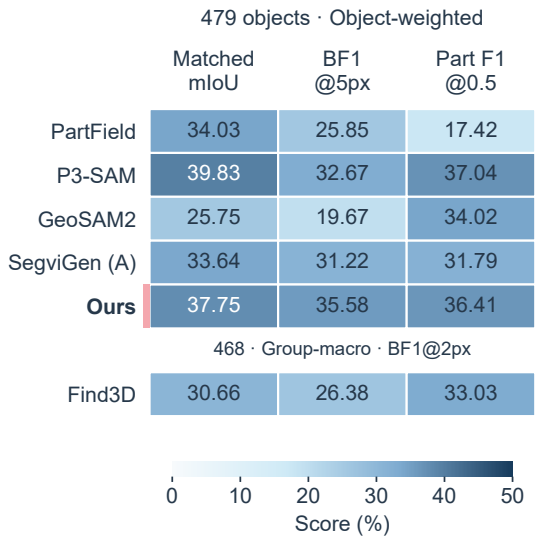


Fig. 4 Metric profiles from Table 1, using one percentage scale without column normalization. The upper block uses 479 objects and BF1@5px; the separate Find3D row uses 468 objects, group-macro weighting, and BF1@2px. Coral marks Ours. The blocks do not support a paired ranking.

4.6.1 Source-View Guidance Quality

Table 2 evaluates the upstream guidance stage after foreground-based frame alignment (`aligned_k`). C reaches 53.64% matched mIoU on 479 objects; B reaches 45.86% on its 440 available objects. The descriptive mean difference is 7.78 percentage points. At five- and two-pixel tolerances, the BF1 differences are 4.88 and 2.80 points, respectively. These differences compare unequal coverage and do not estimate paired improvements. The tighter tolerance preserves the ordering of the available-set means.

Figure 6 compares 2D guidance examples across the four conditions, covering assemblies, solid-volume objects, and thin or fine-grained structures. In the traffic-cone example, the automatic and reference masks are identical. Elsewhere, equal region counts can still conceal different boundaries or ownership. The panels depict the conditions supplied before 3D inference.

4.6.2 Final-Output Projection Agreement

In Table 3, automatic regions raise projection mIoU from 33.64% under A to 35.50% under B. The proposed guidance raises it again to 37.75%

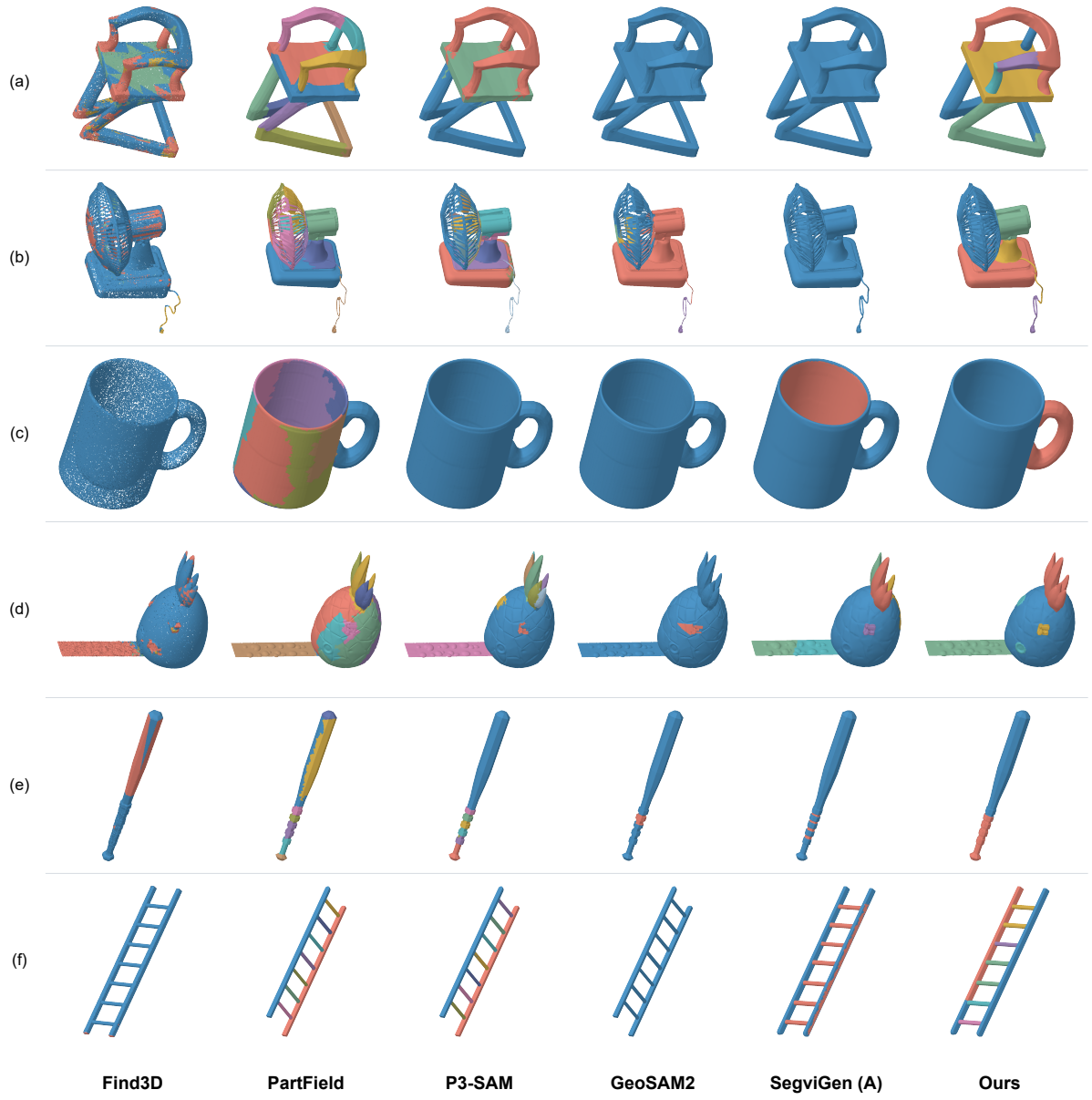


Fig. 5 Six selected favorable cases, two per structural group: (a) chair, (b) fan, (c) mug, (d) pineapple-shaped object, (e) bat, and (f) ladder. Find3D uses its native labeled points; the other methods use labeled meshes. Recorded source-view orientations and uniform display framing are used, with a shared display roll in (f). Colors identify local regions, not shared semantic classes.

Table 1 Final-output projection agreement (percentages; Part F1 at IoU 0.5). Five methods share 479 objects and object weighting. BF1@2px covers 471 objects for the four marked rows; other metrics retain 479. Find3D uses a separate 468-object, group-macro summary and has only verified 2px BF1. Dashes denote unavailable values.

Method	Part condition	Proj. mIoU (%) ↑	BF1@5px (%) ↑	BF1@2px (%) ↑	Part F1@0.5 (%) ↑	Evaluable N
Find3D (Z. Ma et al., 2025)	Text queries	30.66	—	26.38	33.03	468
PartField (Liu et al., 2025)	Feature clustering (K=10)	34.03	25.85	13.98†	17.42	479
P3-SAM (C. Ma et al., 2025)	Automatic	39.83	32.67	18.17†	37.04	479
GeoSAM2 (Deng et al., 2026)	Shared prompts	25.75	19.67	11.40†	34.02	479
SegviGen (Li et al., 2026)	No part prompt	33.64	31.22	—	31.79	479
Ours	2D guidance	37.75	35.58	23.07†	36.41	479

† BF1@2px: $N = 471$; all other entries use the row's stated N.

Table 2 Direct 2D guidance agreement after foreground-based frame alignment, with object-weighted percentages and Part F1 at IoU 0.5. BF1 tolerances share predictions and references within each row. B and C have unequal coverage; mean differences are unpaired percentage points. A has no part map and D is the reference itself.

Condition	Guidance source	N	2D mIoU (%) ↑	BF1@5px (%) ↑	BF1@2px (%) ↑	Part F1@0.5 (%) ↑
A	Raw source image (no part map)	—	—	—	—	—
B	Automatic SAM2 regions	440	45.86	50.44	45.74	40.84
C	Ours	479	53.64	55.32	48.54	44.84
D	Reference map	—	—	—	—	—
Mean difference C−B (unpaired)		—	+7.78	+4.88	+2.80	+4.00

under C and improves BF1 over B by 3.13 points. Because the backend and geometry are shared, the output differences arise from the supplied image condition within this protocol. D reaches 51.30% mIoU, 13.55 points above C, showing that the backend remains sensitive to the input partition. Since D receives the scoring reference as guidance, this value measures reference-conditioned consistency rather than independently measured human-level 3D segmentation. Figure 7 summarizes the comparison.

Figure 8 shows six selected examples rendered from final A/B/C/D outputs. All conditions in a row share the same display view, scale convention, and lighting. C separates the traffic cone from its base and the chair backrest from the seat, whereas B largely merges these structures. C still leaves a small extra region at the cone rim, and several chair legs remain merged with the seat. The examples thus show useful semantic separation without complete recovery of every part. Quantitative scores use the frozen scoring cameras rather than these display views.

4.7 Ablation Of Prompting, Audit, And Assembly

The ablation varies prompts, audit behavior, and assembly while keeping the planning response fixed whenever the removed component does not affect planning. Automatic SAM2 is the image-only baseline. The earlier box-and-point pipeline uses one positive point, no negative points, and no audit. Its assembly gives larger masks overlap priority and fills gaps through repeated 5×5 grayscale dilation. It also omits the proposed component cleaning, small-box priority, and distance-transform filling. Because several operations change together, this row is an earlier-pipeline baseline rather than a single-factor ablation.

The single-positive variant keeps the box, negative points, audit, and assembly but uses only the first positive point in both initial and retry calls. The no-negative variant removes negative points from both calls, and the no-audit variant skips feedback. The no-point-filter variant bypasses the two point-anchored component-filtering passes

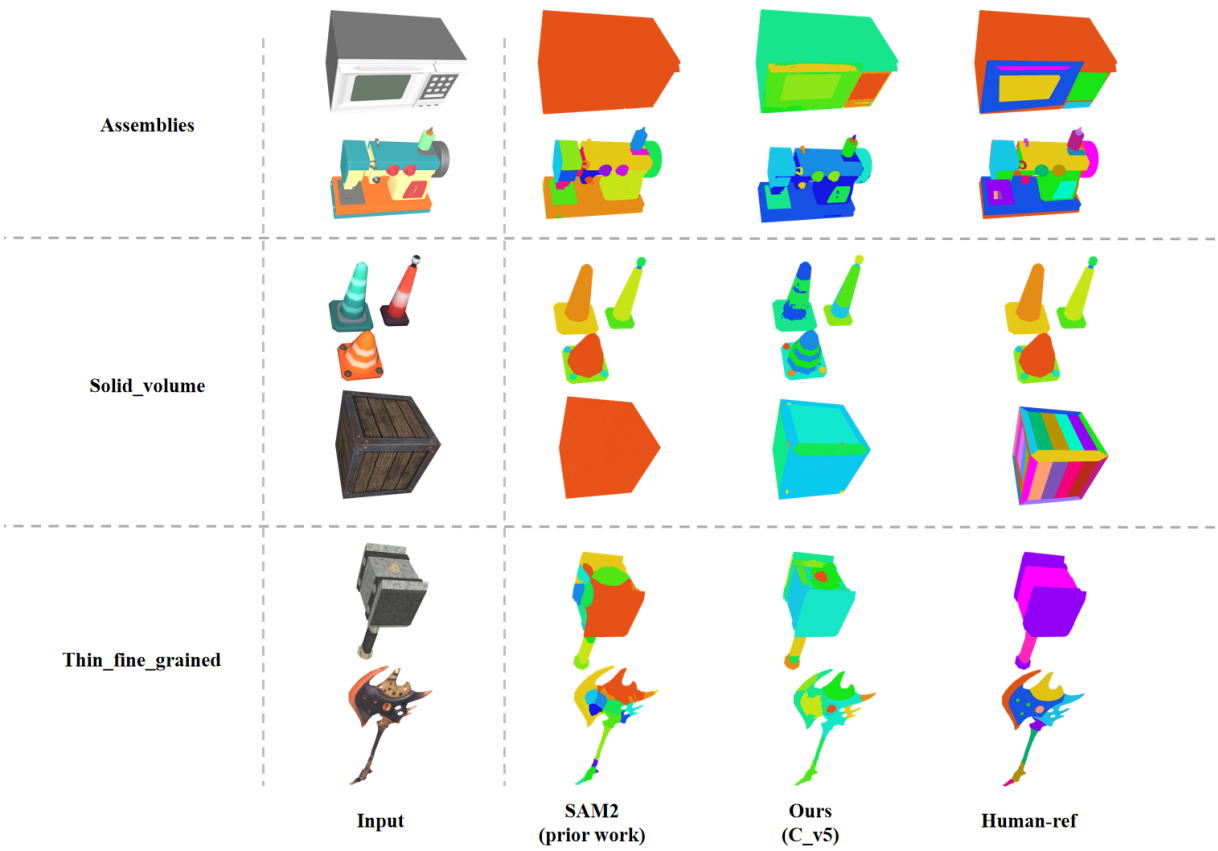


Fig. 6 Source-view guidance examples: input, automatic SAM2, Ours, and human reference, grouped by structural category. Colors encode local regions, not shared semantic IDs. The automatic and reference traffic-cone masks are identical in the supplied data.

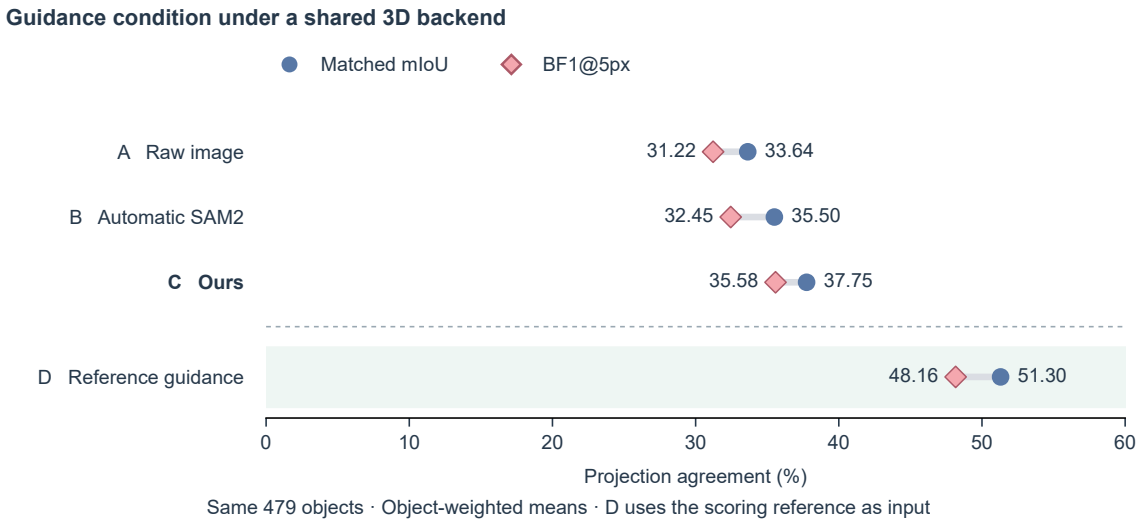


Fig. 7 Controlled projection results from Table 3: circles denote matched mIoU; diamonds denote BF1@5px. Lines connect metrics, not uncertainty intervals. Means use the same 479 objects. The separate D row receives the scoring reference as guidance.



Fig. 8 SegviGen outputs under A/B/C/D guidance, rendered from 24 corresponding GLBs. Each row shares a display view and lighting, independent of the scoring cameras. Rows (a), (b), (d), and (e) are thin/fine-grained objects; (c) is an assembly; (f) is solid volume. Colors identify local output regions.

Table 3 Controlled A/B/C/D projection agreement on 479 objects with shared geometry, SegviGen settings, cameras, and object weights. BF1 uses five pixels at 1024×1024. D receives the scoring reference as guidance and measures reference-conditioned consistency.

Condition	Guidance Input	Objects	Proj. mIoU (%) ↑	Proj. BF1@5px (%) ↑
A	Source image without a part map	479	33.64	31.22
B	Automatic SAM2 regions	479	35.50	32.45
C	Ours, assembled guidance	479	37.75	35.58
D	Designated reference map	479	51.30	48.16

while retaining morphological closing, hole filling, area thresholds, small-box priority, and distance-transform filling. Table 4 reports the available 2D measurements. A paired projection summary is not yet available for these variants, so the evidence is limited to guidance construction.

Removing negative points or point-anchored filtering produces the largest differences from the full configuration: 6.33 and 8.05 mIoU points, respectively. Restricting the prompt to one positive point lowers mIoU by 1.72 points, and removing mask audit lowers it by 1.40 points. Figure 9 displays these aggregate differences without uncertainty intervals. Equal sample counts do not make the comparison paired. *[Confirm identical UID, planning-cache, and D-reference hashes for the updated full row and all four component-removal rows before treating these differences as isolated component effects.]*

4.8 Feedback Behavior On The Archived Cohort

The archived 500-object analysis tests the effect of one-pass correction on the candidates retained by production assembly, rather than rerunning the pipeline without audit. It reconstructs pre- and post-feedback maps from the same candidate list retained by production assembly. The pre-feedback version uses each candidate's initial mask; the post-feedback version substitutes a valid retry when one exists. Retained prompt records cannot reconstruct candidates discarded by production fusion, skipped as hidden, or missing an initial mask, so neither version includes them. The reconstructed post-feedback labels match the stored production labels pixel for pixel on all 500 objects. Table 5 therefore measures retry effects only among candidates that survived production assembly.

The trace contains 4742 planned parts and 4627 valid initial masks. Assembly retained 3546 candidates and discarded 1081. The retained set comprises 2819 direct acceptances, 716 valid retries, five unparseable audits, and six failed retries that kept the initial mask. Audit outcomes were not preserved for discarded candidates, so the difference between all valid initial masks and direct acceptances is not the rejection count. The 115 proposals without a valid initial mask are also separate from audit rejection.

Reconstructed feedback changes macro-averaged mIoU by +0.31 points, part recall by +0.64 points, and Part F1 by −0.05 points. Among the 283 objects with an effective retry, matched mIoU improves for 138, declines for 139, and is unchanged for six. The other 217 objects have identical pre- and post-feedback maps. The thin or fine-grained group has the largest group-level mIoU change at +0.80 points. The small positive mean masks nearly balanced numbers of improvements and declines, so object-level gains are not monotonic. Because the retained logs omit planning and audit wall-clock times, we report calls and outcomes without estimating latency.

4.9 Structural Variation And Failure Cases

The aggregate comparisons do not imply equal performance across structures. We first examine C guidance on the primary 479-object set, then final-output groups on each method's available set, and finally part count and size on the earlier paired 500-object manifest. Each analysis retains its stated cohort and aggregation.

4.9.1 Guidance By Structural Group

Table 6 shows that C guidance has its lowest matched mIoU on assemblies (45.39%), compared

Table 4 2D component configurations against the designated D reference; percentages use equal weighting over three structural groups. Part F1 uses IoU 0.5. Undaggered BF1 is reported at five pixels; † marks unverified five-pixel provenance and prevents direct BF1@5px comparison. Differences are descriptive pending UID and reference reconciliation.

Configuration	Objects	2D Matched mIoU (%) ↑	2D BF1 (%) ↑	2D Part F1 at 0.5 (%) ↑
Automatic SAM2	494	45.33	50.16	40.77
Box-And-Point Baseline	484	43.55	50.02	39.69
Ours With One Positive Point	500	51.79	47.04†	43.58
Ours Without Negative Points	500	47.18	43.24†	40.90
Ours Without Mask Audit	500	52.11	47.53†	44.20
Ours Without Point-Anchored Filtering	500	45.46	40.27†	38.93
Ours	500	53.51	55.35	45.32

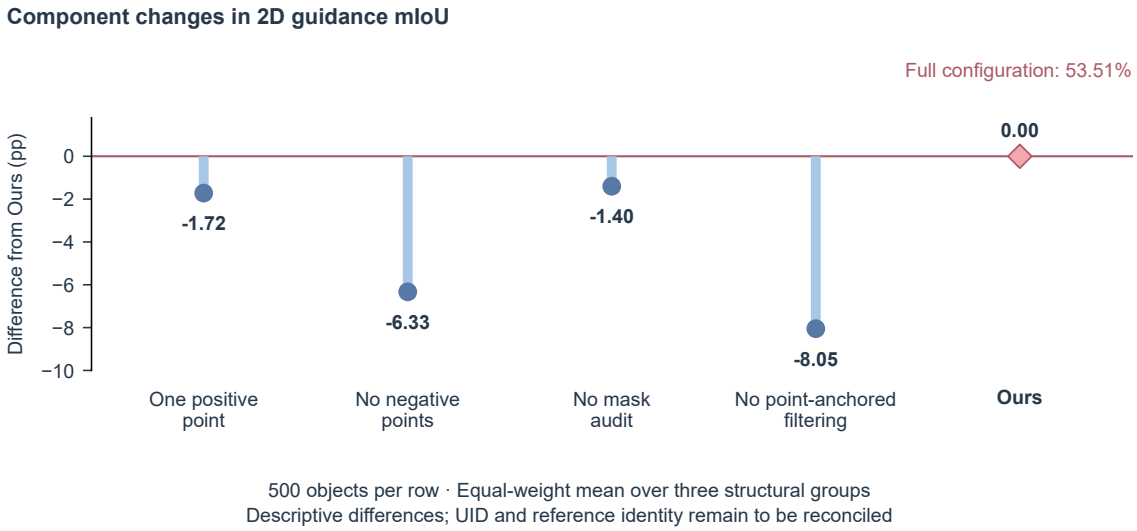


Fig. 9 Descriptive mIoU differences from Ours (53.51%), in percentage points. All five configurations report 500 objects with group-equal weighting. Stems show mean differences, without uncertainty intervals. UID and reference reconciliation is still required before attributing differences to individual components.

Table 5 Feedback on 500 archived objects using the same production-retained candidates. Percentages are three-group macro-averages against the designated D reference; differences use unrounded values. Recall and Part F1 use IoU 0.5.

Guidance	mIoU	Recall	Part F1
Initial masks	51.21	51.06	43.77
Selected masks	51.53	51.70	43.72
Δ (points)	+0.31	+0.64	−0.05

Table 6 C-condition guidance by structural group on 479 objects. Percentages use object weighting, BF1@5px, and Part F1 at IoU 0.5 against the designated reference.

Group	N	mIoU	BF1	Part F1
Assemblies	151	45.39	45.55	45.54
Solid volume	179	56.18	55.56	37.58
Thin / fine-grained	149	58.95	64.94	52.85

with 56.18% on solid-volume objects and 58.95% on thin or fine-grained objects. Despite the narrow support of some components, the thin group has high source-view guidance agreement on this roster. The observed ordering also depends on reference granularity, visible part area, and the retained objects, not just the structural-group label.

4.9.2 Projection By Structural Group

Table 7 gives a descriptive breakdown over each method's available set. In the 500-object summaries, Ours has lower assembly mIoU than P3-SAM and PartField but higher boundary scores than P3-SAM on solid-volume and thin objects. SegviGen's group summaries use 479 objects, so differences from the 500-object Ours rows are unpaired. Table 1 remains the paired overall comparison. The group results show that performance changes are not uniform across geometric structures.

4.9.3 Part Count And Part Size On The Archived Cohort

Table 8 compares B and C on the earlier 500-object manifest: 154 assemblies, 181 solid-volume objects, and 165 thin or fine-grained objects. This set differs in UID membership from the final roster. Scores describe consistency with the designated reference, whose readiness was not independently verified for this analysis. BF1 uses the archived two-pixel tolerance; these values are not five-pixel measurements.

Strata are assigned after fixing the whole-object match, without rematching parts within each bin. Object-count bins use the number of reference parts: 1–5, 6–10, and at least 11. Part-size bins use reference area as a fraction of object foreground: small ($\leq 1\%$), medium ($> 1\%$ and $\leq 5\%$), and large ($> 5\%$). Each object contributes its own mean, followed by equal weighting over the three structural groups. N counts objects contributing to a stratum; size bins may share objects. If a group has no contributing object, the three-group macro-average is left undefined. These are source-view guidance scores and do not require the 3D projection gate.

The three-group macro-averaged matched-mIoU gain is 6.74 points. The largest group-level gain occurs on thin and fine-grained objects, where recall and boundary agreement also improve. Solid-volume objects gain mainly in region overlap, with a small decline in BF1. The group summaries do not identify which individual errors changed. A larger number of predicted regions is not treated as an improvement on its own.

Gains persist in the six-to-ten-part bin, but reverse for objects with at least eleven reference parts. In assemblies, C is 1.56 mIoU points, 3.36 BF1 points, and 1.87 recall points below B. In the thin group, mIoU is 1.93 points lower. The solid-volume group has no objects in this bin, so the stipulated three-group macro-average is undefined. These reversals bound the overall gain without identifying a single cause.

Medium parts occupying 1–5% of foreground area improve by 21.44 mIoU points, compared with 2.90 points for parts occupying at most 1%. Both conditions recover the smallest parts poorly. Area alone cannot explain these differences: neighboring components can compete for ownership, large housing regions can dominate solid-volume objects, and thin structures have limited pixel support. Reference granularity and visible part support therefore remain relevant when interpreting the aggregate improvement.

5 Discussion

The controlled results locate a practical source of variation at the input to a frozen 3D segmenter. Shared geometry and backend settings still permit different part assignments when the source-view condition changes. The contribution is a procedure for constructing that condition, rather than a new 3D representation or segmentation backbone. The external comparison places the improvement in context, because region overlap, boundary agreement, and part recovery favor different methods.

The results suggest complementary roles for prompt exclusion and connected-component support. Negative points discourage inclusion of neighboring parts before mask prediction, while point-anchored filtering removes disconnected spill afterward. Removing either operation changes which pixels a proposed component can claim. The earlier Box-And-Point baseline changes several operations together, so it cannot isolate small-box priority or gap filling. The 2D differences also provide no estimate of the proportional gain after 3D propagation.

Feedback has heterogeneous effects. The archived matched-candidate analysis shows nearly balanced numbers of improved and degraded objects despite a small positive mean change (Table 5). Its interpretation is limited to the observed effect on production-retained candidates. The structural analyses similarly indicate

Table 7 Projection agreement by structural group on each method's available set (N), so cross-method rows are unpaired. Values are percentages; BF1 uses five pixels and Part F1 uses IoU 0.5. SegviGen uses condition A.

Structural group	Method	N	Matched mIoU (%) ↑	BF1@5px (%) ↑	Part F1@0.5 (%) ↑
Assemblies	P3-SAM	157	30.25	24.60	26.90
	GeoSAM2	157	12.80	9.18	15.90
	PartField	157	30.17	21.38	23.77
	SegviGen (A)	151	15.22	13.54	14.28
	Ours	157	24.80	22.12	25.35
Solid volume	P3-SAM	181	47.25	31.03	46.17
	GeoSAM2	181	41.04	28.52	52.34
	PartField	181	35.39	19.30	10.83
	SegviGen (A)	179	45.84	36.68	39.97
	Ours	181	46.69	38.73	42.79
Thin / fine-grained	P3-SAM	162	41.06	43.37	37.25
	GeoSAM2	162	20.69	19.99	30.05
	PartField	162	36.47	38.56	19.40
	SegviGen (A)	149	37.65	42.57	39.72
	Ours	162	41.08	46.73	41.07
All	SegviGen (A)	479	33.64	31.22	31.79

Table 8 2D structural and part-size analysis on the archived 500-object B/C paired set. Scores are percentages; differences are C minus B in percentage points. BF1 uses two pixels; recall uses IoU 0.5. N counts contributing objects. Group rows use object weighting; part-count, part-size, and overall rows weight the three groups equally. An empty group leaves the macro-average undefined.

Panel	Stratum	N	Matched mIoU			BF1@2px			Recall@0.5		
			B	C	Δ	B	C	Δ	B	C	Δ
Group	Assemblies	154	37.12	42.60	+5.48	34.80	34.62	−0.19	38.49	44.57	+6.08
	Solid volume	181	50.50	56.35	+5.85	49.42	49.23	−0.19	51.15	52.15	+1.00
	Thin / fine-grained	165	46.75	55.63	+8.88	49.98	56.75	+6.76	48.75	58.38	+9.63
Part count	1–5 parts	327	53.49	61.92	+8.43	54.55	56.57	+2.01	55.61	62.89	+7.28
	6–10 parts	107	34.91	44.47	+9.55	31.90	37.81	+5.91	35.15	46.63	+11.48
	≥11 parts	66	—	—	—	—	—	—	—	—	—
Part size	Small (≤1%)	158	13.38	16.28	+2.90	20.27	20.92	+0.65	11.64	14.25	+2.61
	Medium (1–5%)	261	23.87	45.31	+21.44	32.50	45.19	+12.69	22.12	47.20	+25.07
	Large (>5%)	500	54.01	59.06	+5.05	49.98	50.73	+0.75	56.42	59.54	+3.12
Overall	All objects	500	44.79	51.53	+6.74	44.74	46.87	+2.13	46.13	51.70	+5.57

that a useful partition depends on reference granularity, part size, and the visible support available to the prompts.

Evaluation is limited to agreement with the designated visible-region reference under the recorded source camera. It does not establish segmentation accuracy on unobserved surfaces. D conditions the backend on that same reference, so its score measures reference-conditioned consistency. The retained collection is also conditional on asset availability and successful C runs. Independent multiview part annotations would allow a broader test of whether the improved source-view partition remains useful beyond the observed view.

6 Conclusion

We presented a language-guided spatial prompting pipeline for frozen 3D segmentation. Named

boxes and points specify visible parts, overlay audit allows one correction, and deterministic assembly produces complete guidance. On the controlled 479-object projection set, guidance improves matched mIoU over automatic SAM2 by 2.25 points without retraining the backend. Direct 2D available-set means also favor C, although B covers fewer objects. Component summaries identify negative points and point-anchored filtering as promising contributors; their isolated effects still require paired asset reconciliation. The results establish source-view reference agreement, leaving unobserved-surface accuracy for evaluation with independent multiview annotations.

Statements And Declarations

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Competing Interests

The authors declare that they have no competing interests.

Author Contributions

Yulin Fu led the study, including its conception, methodology development, experiments, and initial manuscript draft. Cheng Yang and Qing Ai assisted with comparative experiments, validation, and verification of experimental settings. Qibin Hu contributed to discussions of the research ideas and feasibility analysis. Bin Hu and Jisheng Dang supervised the research, coordinated the project, and contributed to manuscript revision. All authors reviewed and approved the final manuscript.

Ethics Approval

This study used publicly available datasets and did not require ethics approval.

Data Availability

The data generated and analyzed in this study are not currently publicly available.

Code Availability

[GitHub repository URL, frozen configuration, environment, and evaluation scripts.]

Use Of Generative Tools

The authors used OpenAI Codex and Google Gemini to check and supplement the experimental design and to assist with language editing. The authors verified the resulting changes and remain responsible for the experimental design, analysis, and manuscript.

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